

Carpool Marketplace Payment System Design

Goals

Marketplace payments, escrow-like flow, payouts, reliability and accounting.

Architecture

Rider App → Payment API → Stripe → Webhooks → Queue/Outbox → Payment Processor → Ledger → Payout Service → Stripe Connect → Driver Bank.

Services

Payment Service, Ledger Service, Payout Service, Webhook Handler, Reconciliation Service, Dispute Service, Wallet Service, Outbox/Event Processor.

Stripe Connect

Use Express accounts with Stripe-hosted onboarding and KYC.

Payment Lifecycle

Create PaymentIntent → Payment Success → Held in Escrow → Booking Completed → Payout Eligible → Transfer → Driver Bank.

Queues & Events

Use Outbox pattern and event-driven processing for retries and resiliency.

Reconciliation

Hourly incremental checks and daily settlement reconciliation against Stripe.

Ledger

Append-only accounting entries, avoid mutable balances.

Edge Cases

Duplicate webhooks, payout failures, refunds, disputes, server downtime.

Architecture Flow

Rider → Payment API → Stripe → Webhook Handler → Event Queue → Payment Processor → Ledger/Payout → Stripe Connect → Driver Bank